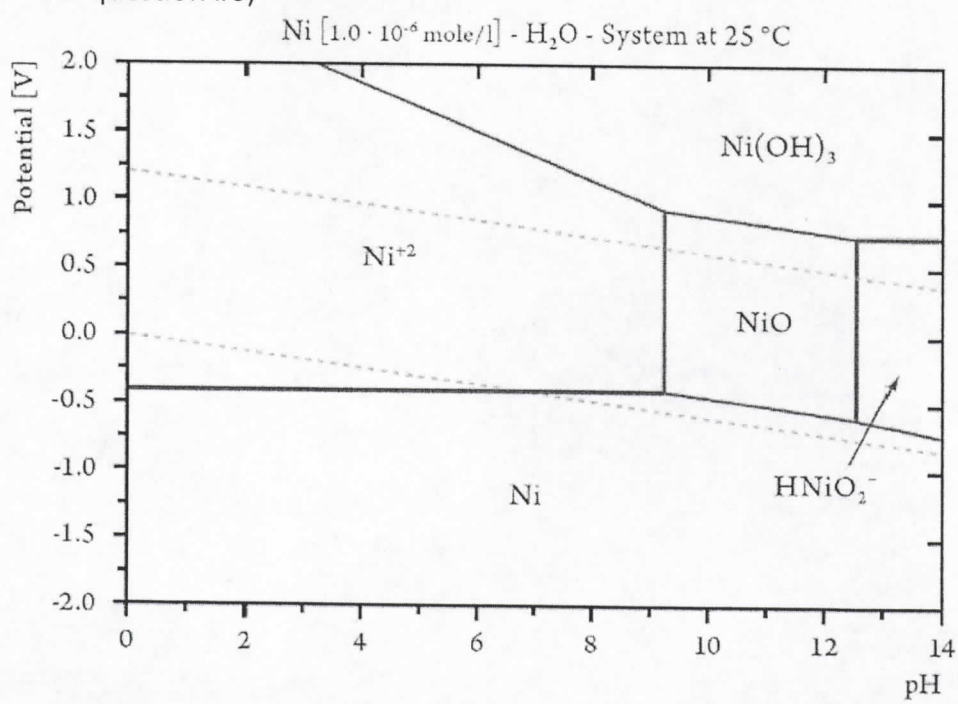
Inorganic Cumulative Exam
Feb 19 2016

Answer any 5 of the following 10 questions (20 points each). Any answers provided beyond the requested five will not be graded.

1. Consider the general reduction reaction of the complex cobalt ion $[Co(H_2O)_6]^{3+}$ to $[Co(H_2O)_6]^{2+}$.
   a. (10 pts) Would you expect this reaction to proceed through an outer-sphere path or an inner-sphere path? Why?
   b. (10 pts) What structural changes would you expect (if any) to occur as a result of the reduction? Give an explanation for your expectations.
2. Materials like graphene and transition metal dichalcogenides (WSe_2, MoS_2, etc.) are completely stable as a monolayer. They are also found as bulk, three dimensional crystals (layered graphene is graphite, for example). Explain the inter-sheet interactions/bonding that are found in these three dimensional structures.
3. Cu(II) compounds (d^9 complexes like $[Cu(H_2O)_6]^{2+}$) could be simply assumed to have a single absorption band in a UV-vis measurement. However, two distinct, overlapping peaks are observed. Why is this?
4. Define the photoelectric effect. How can it be used to identify the chemical state (oxidation state) of an element in a molecule or species?
5. Cl_2 is a non-polar molecule. When it is cooled below -100C, it forms a solid without forming new covalent bonds. Describe how you view the intermolecular interactions that result in an organized solid.

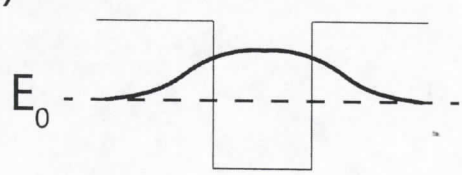
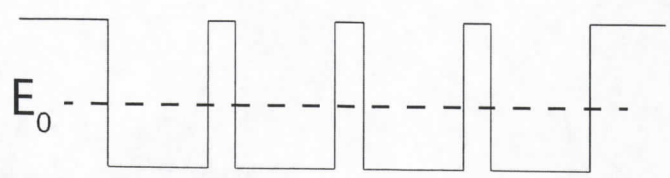
(Figure for question #6)



6. For the following questions, refer to the Pourbaix diagram for a dilute aqueous solution of Nickel (above). The y-axis (Potential [V]) is measured with respect to the Normal Hydrogen Electrode (NHE) potential.
    a. (5pts) What do the dashed lines represent?
    b. (5 pts) Under what potentials and pH is NiO thermodynamically stable?
    c. (5 pts) Under what conditions can a Ni electrode be dissolved in solution?
    d. (5 pts) In what approximate pH range is Ni thermodynamically stable in a solid phase regardless of electrochemical potential?
7. How is the x-ray diffraction pattern of a powder of a given crystal different than the x-ray diffraction pattern of a single crystal? Why is it different?
8. (a, 15pts) From the perspective of electronic bonding, describe how semiconductors like TiO_2 or ZnO can be intrinsically doped by simply removing a small fraction of oxygen atoms from the lattice. (b, 5pts) Is the intrinsically doped TiO_2 n-type or p-type?
9. Ferrocyanide, $[Fe(CN)_6]^{4-}$, does not have an absorption band in the visible spectrum. Ferricyanide, $[Fe(CN)_6]^{3-}$, absorbs strongly at ~400nm, resulting in red crystals in the solid phase. In a few sentences, describe the type of electron transition that is responsible for this strong absorption.
10. A simple derivation of one-dimensional band theory involves building exceedingly longer chains of identical potential wells. In the figure below (a) shows the ground state, s-like electronic wavefunction of a single finite square well potential and its ground state energy E_0. Using this graphical result, draw the unique 'linear combinations of atomic orbitals' that represent the molecular orbitals for the four finite-square well chain (b). Plot the energy of each combination with respect to E_0.

(Figure for question #10)

(a)



(b)